CS471 – Web Technologies (Laboratory)	Qassim University College of Computer	Lab 6
		HTML Form

This lab session covers the basics of HTML form. By the end of this session, students should have a very good understanding of how to build a basic form and how to Handle form in Django.

Pre-lab Preparation:

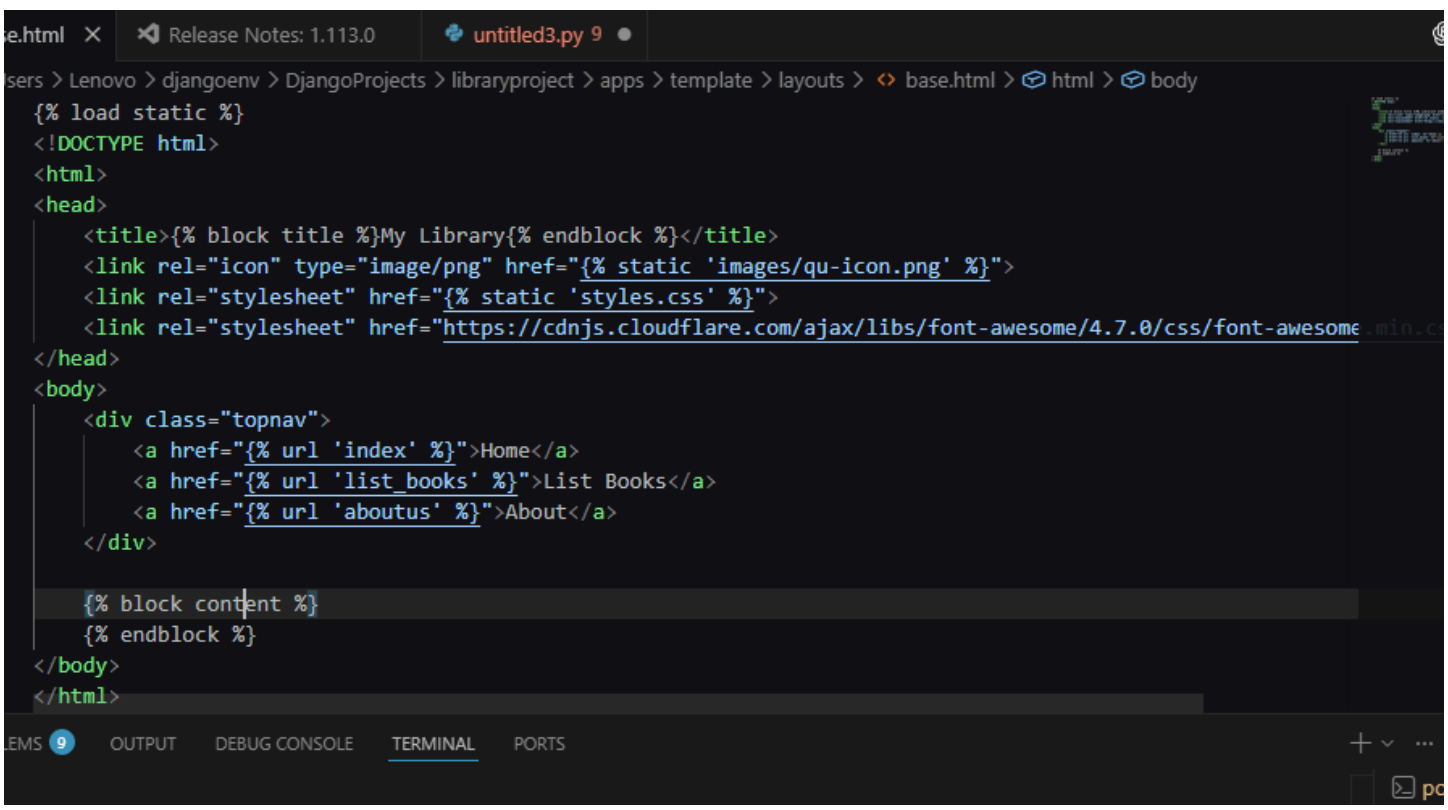
1. Having bookmodule and usermodule in apps directory, each containing separate URL configurations.
2. Template directory located in 'apps' with necessary settings.
3. Master template 'base.html'

Lab Activities: A complete webpage example with html form and Handling form in Django

Task 1: Build HTML form template

Step 1: Using the master template base.html from Lab4, import it in <head> section in base.html the following:

```
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">
```



Step 2: using the master template 'base.html', Create a view function and URL '/books/search' to a page that contains the following HTML form:

```
{% extends "layouts/base.html" %}
{% load static %}
{% block title %} Searching Page {% endblock title %}
{% block stylesheets %}
<style>
    .searchForm{
        margin-top: 20px;
    }

    button {
        color: #fff;
        background-color: #17a2b8;
        border-color: #17a2b8;
        border: none;
        padding: 16px 32px;
        text-align: center;
        text-decoration: none;
        display: inline-block;
        font-size: 16px;
        margin: 4px 2px;
        transition-duration: 0.4s;
        cursor: pointer; /* hand pointer */
    }
    button:hover {
        background-color: green;
        color: white;
    }

</style>
{% endblock stylesheets %}

{% block content %}
<form class="searchForm" method="post" action="">
    {% csrf_token %}
    <label>Keywords:</label>
    <input type="text" name="keyword" placeholder="enter a keyword"/>
    <br/>

    <label>Search in:</label>
    <input type="checkbox" name="option1"/><label>Title</label>
```

```
    <input type="checkbox" name="option2"/><label>Author</label>
    <br/>

    <button type="submit" ><i class="fa fa-search" aria-hidden="true"></i> Search</button>
</form>
{% endblock content %}
```

```
> Users > Lenovo > djangoenv > DjangoProjects > libraryproject > apps > template > bookmodule >
1   {% extends "layouts/base.html" %}
2   {% load static %}
3
4   {% block title %} Searching Page {% endblock title %}
5
6   {% block stylesheets %}
7   <style>
8       .searchForm {
9           margin-top: 20px;
10      }
11
12      button {
13          color: #fff;
14          background-color: #17a2b8;
15          border-color: #17a2b8;
16          border: none;
17          padding: 16px 32px;
18          text-align: center;
19          text-decoration: none;
20          display: inline-block;

OBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

C:\Users\Lenovo>
```

Step 3: After running the server, open the web browser and navigate to http://127.0.0.1:8000/books/search. You should see the form.

base.html | search.html | views.py 2 | Release Notes: 1.113.0

C: > Users > Lenovo > djangoenv > DjangoProjects > libraryproject > apps > bookmodule > views.py > ...

```python
from django.shortcuts import render
from django.http import HttpResponse

# Create your views here.
def search_books(request):
    return render(request, 'bookmodule/search.html')


def index(request):
    return render(request, "bookmodule/index.html")

def list_books(request):
    books = [
        {'title': 'لأنك الله', 'image': 'images/book1.jpg'},
        {'title': 'العادات الذرية', 'image': 'images/book2.jpg'},
        {'title': 'الأب الغني والأب الفقير', 'image': 'images/book3.jpg'},
```

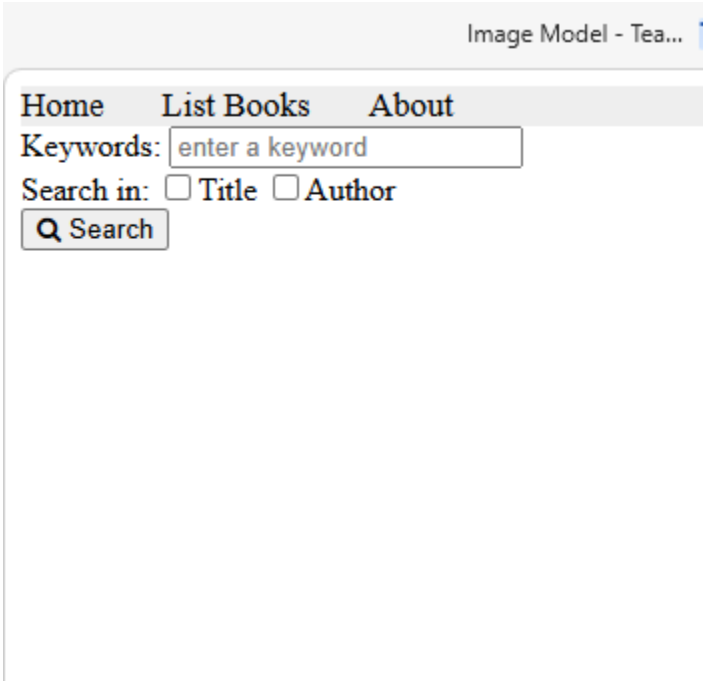
base.html | search.html | views.py 2 | urls.py 1 | Release Notes: 1.113.0

C: > Users > Lenovo > djangoenv > DjangoProjects > libraryproject > apps > bookmodule > urls.py > ...

```python

from django.urls import path
from . import views

urlpatterns = [
    path('search/', views.search_books, name='search_books'),
    path('', views.index, name="index"),
    path('list_books/', views.list_books, name="list_books"),
    path('one_book/', views.one_book, name="one_book"),
    path('aboutus/', views.aboutus, name="aboutus"),
    path('html5/links', views.lab5_links, name='lab5_links'),
    path('html5/text/formatting', views.lab5_formatting, name='lab5_formatting'),
    path('html5/listing', views.lab5_listing, name='lab5_listing'),
    path('html5/tables', views.lab5_tables, name='lab5_tables'),
]
```



Task 2: Handle the form once submitted

Step 1: In 'apps/bookmodule/view.py', create the following function:

```
def __getBooksList():
    book1 = {'id':12344321, 'title':'Continuous Delivery', 'author':'J.Humble and D. Farley'}
    book2 = {'id':56788765,'title':'Reversing: Secrets of Reverse Engineering', 'author':'E. Eilam'}
    book3 = {'id':43211234, 'title':'The Hundred-Page Machine Learning Book', 'author':'Andriy Burkov'}
    return [book1, book2, book3]
```

Step 2: In the same corresponding view/controller in views.py that shows the form, add the code where If there is an action in the form, the form needs to be handled in that view.

Note: it is possible to direct the action to an exclusive view function specific to handle the submitted form.

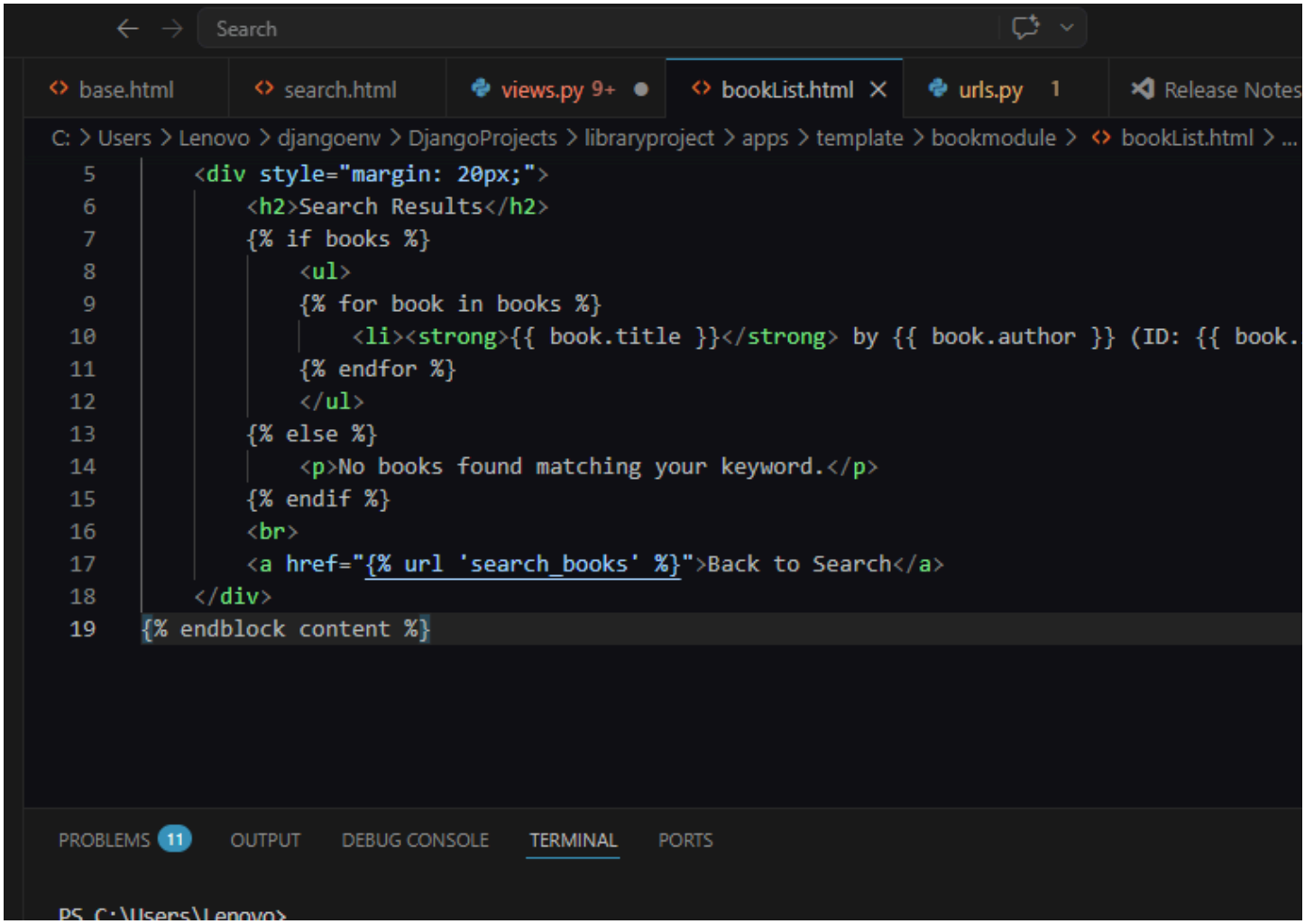
```
if request.method == "POST":
        string = request.POST.get('keyword').lower()
        isTitle = request.POST.get('option1')
        isAuthor = request.POST.get('option2')
        # now filter
        books = __getBooksList()
        newBooks = []
        for item in books:
            contained = False
            if isTitle and string in item['title'].lower(): contained = True
```

```
        if not contained and isAuthor and string in item['author'].lower():contained = True

        if contained: newBooks.append(item)
    return render(request, 'bookmodule/bookList.html', {'books':newBooks})
```

Step 3: Create an HTMl file '`bookList.html`' to list the books (you should use the master template '`base.html`')

Step 4: Now from the web browser, navigate to http://127.0.0.1:8000/books/search, and submit the form with data.



```python
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from django.http import HttpResponse

def __getBooksList():
    book1 = {'id':12344321, 'title':'Continuous Delivery', 'author':'J.Humble and D. Farley
    book2 = {'id':56788765, 'title':'Reversing: Secrets of Reverse Engineering', 'author':'
    book3 = {'id':43211234, 'title':'The Hundred-Page Machine Learning Book', 'author':'And
    return [book1, book2, book3]

def search_books(request):
    if request.method == "POST":
        string = request.POST.get('keyword').lower() # الكلمة المفتاحية [cite: 63]
        isTitle = request.POST.get('option1')        # البحث في العنوان [cite: 64]
        isAuthor = request.POST.get('option2')       # البحث في الكاتب [cite: 65]

        books = __getBooksList() [cite: 67]
        newBooks = [] [cite: 68]

        for item in books: [cite: 69]
            contained = False [cite: 70]
```

```
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\Users\Lenovo>
```

Home List Books About

Keywords: Continuous Delivery

Search in: ☑ Title ☐ Author

Search

Home List Books About

Search Results

- **Continuous Delivery** by J.Humble and D. Farley (ID: 12344321)

Back to Search